COL 703/7203

Nov. 12th

Validity of $\exists x \forall y\, P(x,y) \rightarrow \forall y \exists x\, P(x,y)$

is same as unsatisfiability of

$$\neg\left(\exists x \forall y\, P(x,y) \rightarrow \forall y' \exists x'\, P(x',y')\right)$$

$$\exists x \forall y\, P(x,y) \;\wedge\; \neg \forall y' \exists x'\, P(x',y')$$

$$\wedge\; \exists y' \forall x'\, \neg P(x',y')$$

$$\forall y\, P(a,y) \;\wedge\; \forall x'\, \neg P(x',b)$$

$P(a,a)$ $\quad P(a,b)$ $\quad \neg P(a,b)$ $\quad \neg P(b,b)$

□

Semi-decision procedure for validity of ϕ

i) Negate ϕ

ii) Write it in clausal form

iii) Generate ground instances

iv) Resolve to get $\square$

No efficient way to generate the set of ground instances which would resolve to $\square$.

Exercise

$$\forall x \exists y \left(P(x) \to Q(y) \right) \to \exists y \forall x \left(P(x) \to Q(y) \right)$$

prove that this is valid

$$\neg \left(\forall x \exists y \left(P(x) \to Q(y) \right) \to \exists y \forall x \left(P(x) \to Q(y) \right) \right)$$

same as this only must

$$\forall x \exists y \left(P(x) \to Q(y) \right)$$

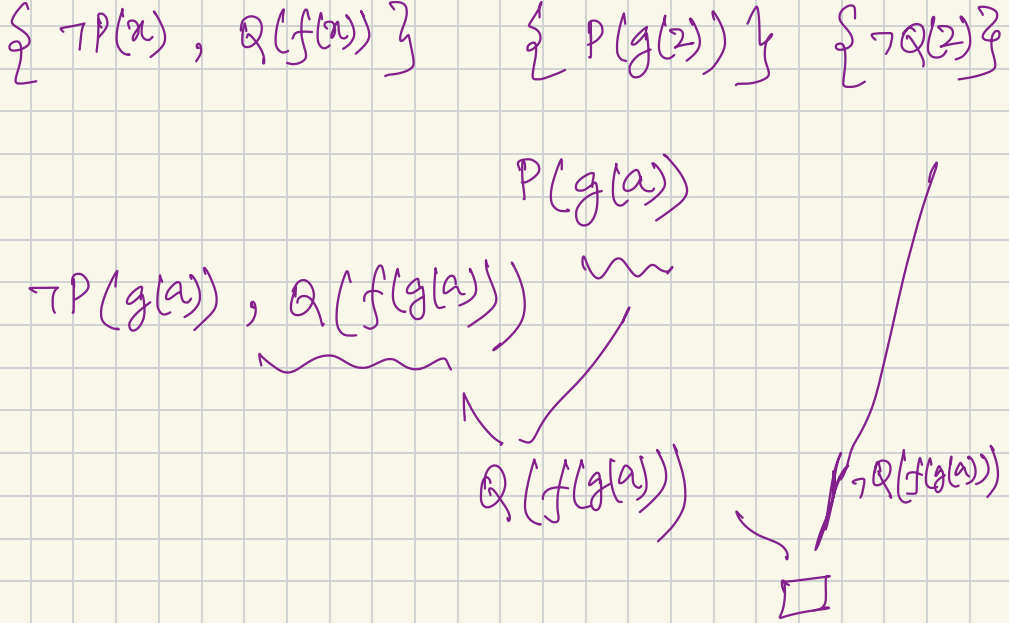
$$\forall y \exists x \neg \left(P(x) \to Q(y) \right)$$

$$\forall x \left(\neg P(x) \vee Q(f(x)) \right)$$

$$\forall z \exists w \neg \left(P(w) \to Q(z) \right)$$

$$\forall z \exists w \left(P(w) \wedge \neg Q(z) \right)$$

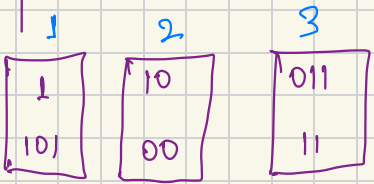
$$\forall z \left(P(g(z)) \wedge \neg Q(z) \right)$$


{ ¬P(x) , Q(f(x)) }
{ P(g(z)) }
{ ¬Q(z) }
P(g(a))
¬P(g(a)) , Q(f(g(a)))
Q(f(g(a)))
¬Q(f(g(a)))

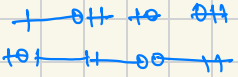
Undecidability of satisfiability and validity

Post's Correspondence Problem (PCP)

Tiles:



1323 is a solution.



A solution to this problem is a sequences of tiles (allowing multiple instances of each tile) such that the string at the top matches the one at the bottom.

Encode this in FO logic

Signature σ

e	constant symbol
f_0, f_1	two function symbols
P	Predicate symbol.

101

$f_1(f_0(f_1(e)))$

$f_{101}(e)$

ϕ_1 $\quad P(f_1(e), f_{101}(e))$

$\wedge\ P(f_{10}(e), f_{00}(e))$

$\wedge\ P(f_{011}(e), f_{11}(e))$

ϕ_3 $\quad \exists u P(u,u)$

ϕ_2 $\quad \forall u \forall v \Big(P(u,v) \rightarrow$

$P(f_1(u), f_{101}(v))$

$\wedge\ P(f_{10}(u), f_{00}(v))$

$\wedge\ P(f_{011}(u), f_{11}(v)) \Big)$

Claim: $\phi_1 \wedge \phi_2 \to \phi_3$ is valid iff the PCP instance has a solution.

Suppose $\phi_1 \wedge \phi_2 \to \phi_3$ is valid.

Let us take a Herbrand structure H for which $P_H = \{ (f_u(e), f_v(e)) \mid \exists t_1 \dots t_k \}$

$u = x_{t_1} \dots x_{t_k}$

$v = y_{t_1} \dots y_{t_k}$

$\leftarrow$ $\boxed{x_i \; y_i}$ are the tiles.

H satisfies $\phi_1 \wedge \phi_2$

$\therefore H$ satisfies ϕ_3

which means that the PCP instance has a solution.

Show

Assume the PCP instance has a soln.

We need to prove that $\phi_1 \wedge \phi_2 \rightarrow \phi_3$ is valid.

Consider any structure A that satisfies ϕ_1 and ϕ_2.

by induction on k.

For any sequence of tiles $t_1 \ldots t_k$

$$A \models P(f_u(e), f_v(e))$$

where $u = x_{t_1} \ldots x_{t_k}$

$v = y_{t_1} \ldots y_{t_k}$

Because the PCP instance has a solution

there is some $t_1 \cdots t_n$

$u = x_{t_1} \cdots x_{t_n}$

$u = y_{t_1} \cdots y_{t_n}$

$A \models P(f_u(e), f_u(e))$

$A \models \exists u \, P(u, u)$

$A \models \phi_3$.

Validity is semi-decidable ←

" " undecidable

Sat is undecidable

Sat is *not* semi-decidable.